DSCI 510

Weather Effects on Urban Air Pollution

Los Angeles vs Houston

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Weather vs PM2.5 in two cities

This project analyzes how weather variables — temperature, humidity, wind speed, and precipitation — affect daily PM2.5 in Los Angeles and Houston from January – June 2024.

PM2.5 is a leading cause of pollution-related health problems, and its daily level varies with weather. LA (coastal Mediterranean) and Houston (humid subtropical) sit in very different climates, so the same variable may drive pollution differently in each.

Q1

Do the same weather variables drive PM2.5 in Los Angeles and Houston?

Q2

Does a nonlinear model (Random Forest) beat a linear baseline at predicting PM2.5?

Three independent sources

#	Name / short description	Source URL	Type	List of fields	Format	Data size
1	**EPA Air Quality System (AQS) API.** Daily PM2.5 and AQI by county	aqs.epa.gov/aqsweb/documents/data_api.html	API	date_local, arithmetic_mean, aqi, state, county	JSON	Multiple monitor-level records, aggregated to ~182 daily observations per city
2	**Open-Meteo Historical Weather API.** Daily historical weather for selected coordinates	open-meteo.com/en/docs/historical-weather-api	API	time, temperature_2m_mean, relative_humidity_2m_mean, wind_speed_10m_mean, precipitation_sum	JSON	~182 daily observations per city
3	**EPA AirData Historical Dataset.** Daily AQI by county used for cross-source validation	aqs.epa.gov/aqsweb/airdata/download_files.html	File	Date, State Name, county Name, AQI	CSV	National file filtered to Los Angeles County and Harris County

Final analysis dataset: 364 daily rows (2 cities × 182 days) after merging EPA AQS + Open-Meteo on (city, date) and dropping rows with missing values.

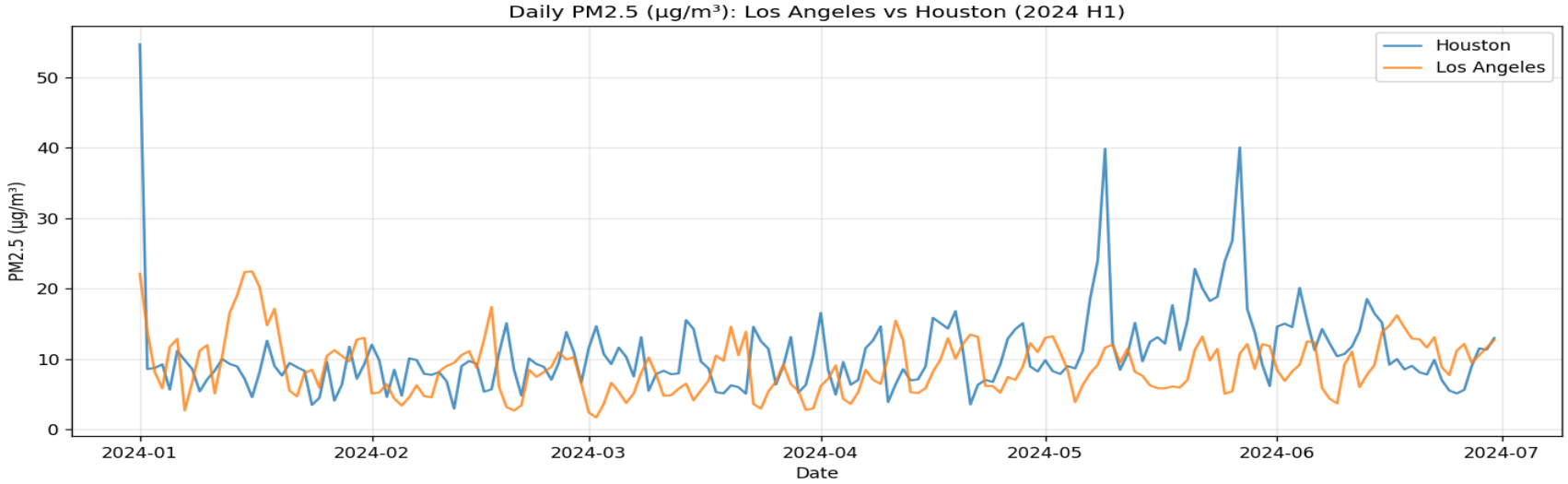
Source #3 (CSV) is independent of the EPA API — it lets us cross-validate the API's AQI values against a second aggregation of the same monitoring network.

Variables used in the analysis

Feature	Description	Unit	Source field
temperature	Daily mean 2 m air temperature.	°C	Open-Meteo temperature_2m_mean
humidity	Daily mean 2 m relative humidity.	%	Open-Meteo relative_humidity_2m_mean
wind_speed	Daily mean 10 m wind speed.	km/h	Open-Meteo wind_speed_10m_mean
precipitation	Daily total precipitation.	mm	Open-Meteo precipitation_sum
pm25 (target)	Daily mean PM2.5 across county monitors.	$\mu g/m^3$	EPA AQS API arithmetic_mean

All four weather features come from Open-Meteo; the target (pm25) comes from the EPA AQS API.

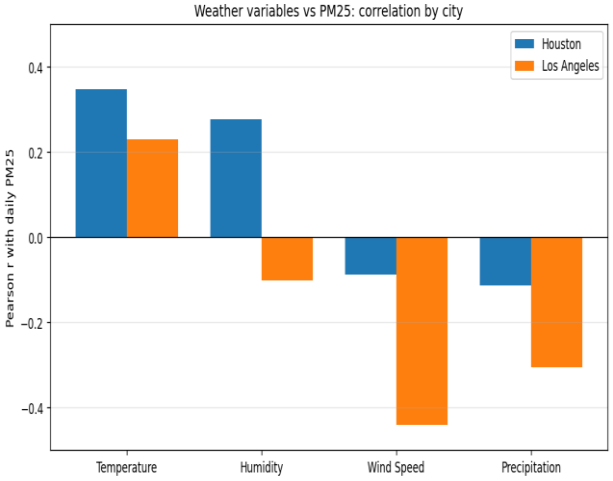
Daily PM2.5 over time

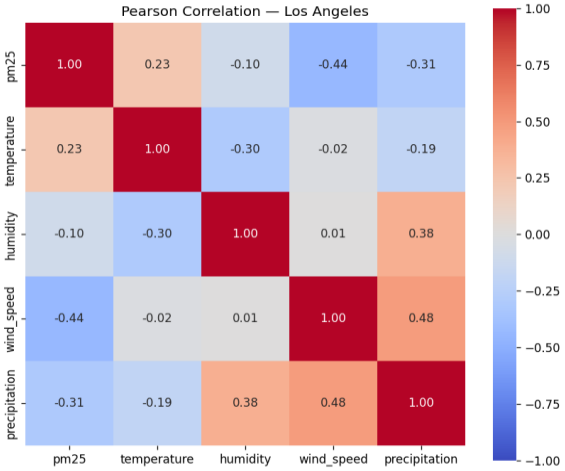


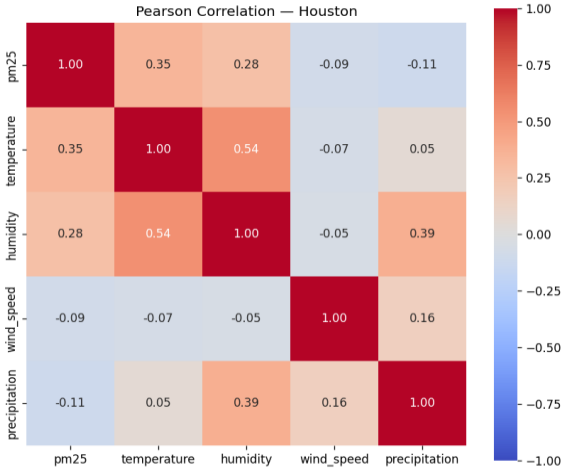
Houston shows extreme PM2.5 spikes (Jan ~55 µg/m³, late May ~40 µg/m³); LA stays comparatively flat.

These spikes likely reflect events weather alone can't explain — supporting why test R^2 caps at ~0.25.

Weather vs PM2.5: correlation by city







Takeaways

- Different top drivers per city — wind in LA (r = −0.44), temperature in Houston (r = +0.35).
- LA full matrix: wind speed is the strongest negative driver; precipitation also reduces PM2.5.
- Houston full matrix: temperature is the strongest positive driver; humidity also positive.

Linear baseline vs Random Forest

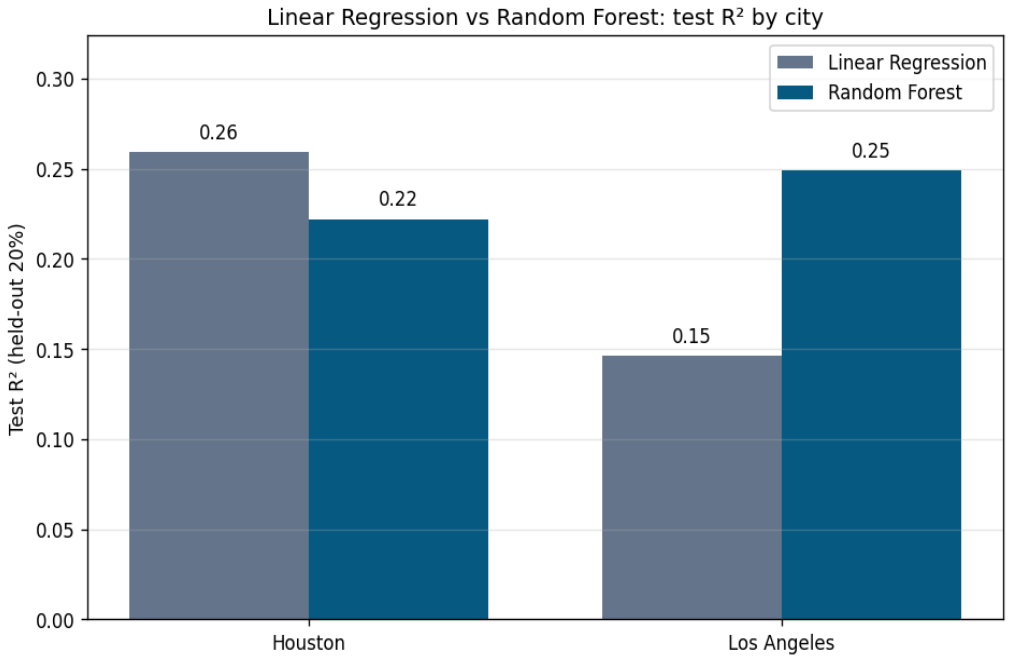
Both models use the same 80/20 train/test split and the same four weather features — so test R^2 and RMSE are directly comparable.

Test R^2 — who wins?

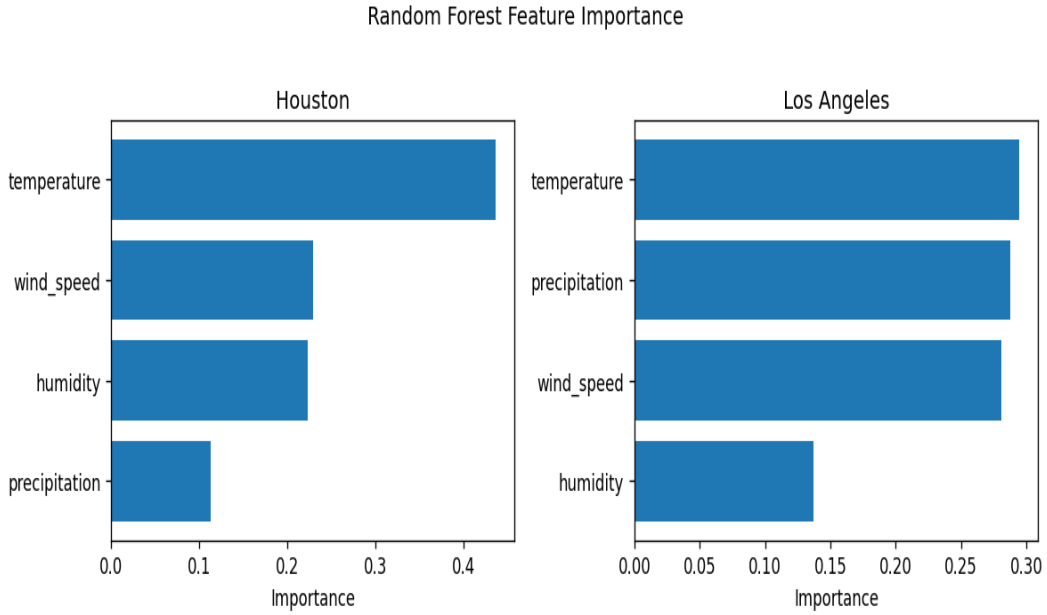
Los Angeles: RF 0.25 > LR 0.15

Houston: LR 0.26 > RF 0.22

Weather explains ~15–25% of daily PM2.5 variation — the rest must come from sources we didn't measure (emissions, regional transport).



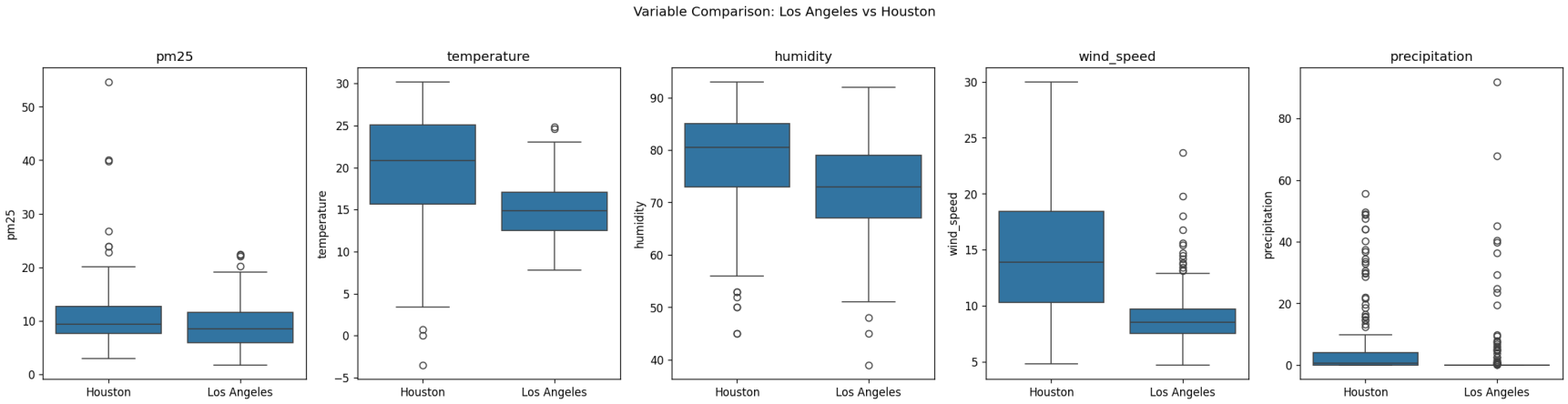
Random Forest feature importance



Insights

- Houston → temperature dominates (~0.44).
- LA → three-way tie: temperature, precipitation, wind (each ~0.28–0.29).
- RF surfaces nonlinear structure — top RF feature ≠ top Pearson-r feature in LA.

Distributions by city: 5 variables



LA vs Houston differ on every variable — wider PM2.5 spread in Houston, very different temperature / humidity / wind regimes.

Houston

wider PM2.5 range and more outliers, hotter and more humid.

Los Angeles

tighter PM2.5 distribution; cooler, drier, windier.

Why it matters

different feature distributions → different model behavior per city.

Cross-source validation

Merging the EPA AQS API AQI with the independently aggregated EPA AirData CSV on (city, date) and correlating the two lets us check the analysis against a second source.

r = 0.60

Los Angeles

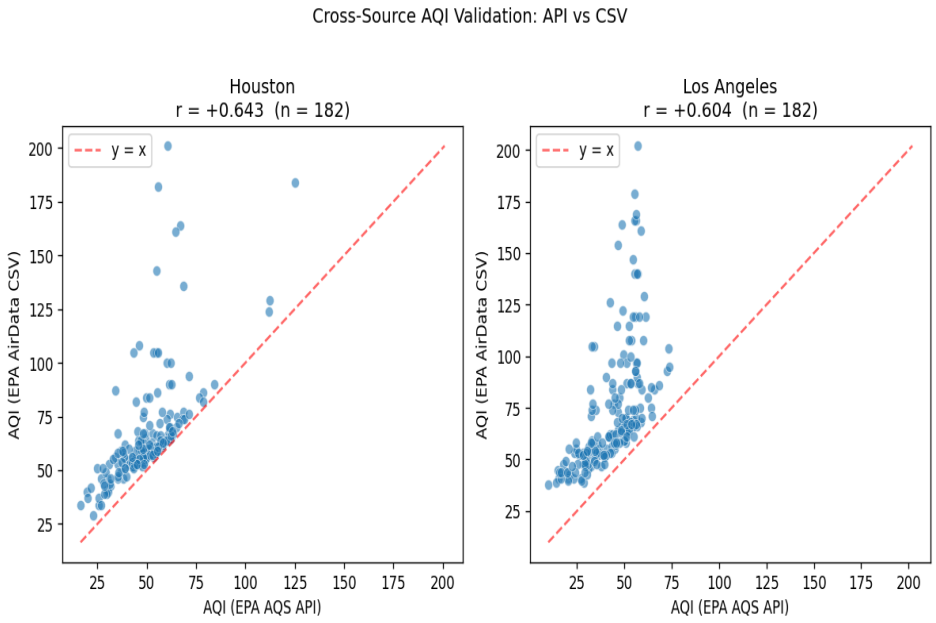
n = 182

r = 0.64

Houston

n = 182

Sources track together (r ≈ 0.6) but disagree on level — the CSV's AQI runs systematically higher, likely because it uses worst-pollutant AQI across more monitors.



What was hardest

1 EPA AQS API: rate limits & credentials

The EPA API limited long requests, and invalid or expired credentials caused failures. We had to batch requests by state, county, and month, and rebuild the pipeline after the registered key expired. A failure at this step can return an empty DataFrame and break downstream joins.

2 Weather alone explains only 15–25% of PM2.5

Test R^2 values of about 0.15 to 0.26 show that weather variables explain only part of the daily PM2.5 variation. Important factors such as emissions, regional transport, and wildfire smoke were not included in this project. This limited the predictive power of the models and required more careful interpretation of the results.

Thank you

Questions?